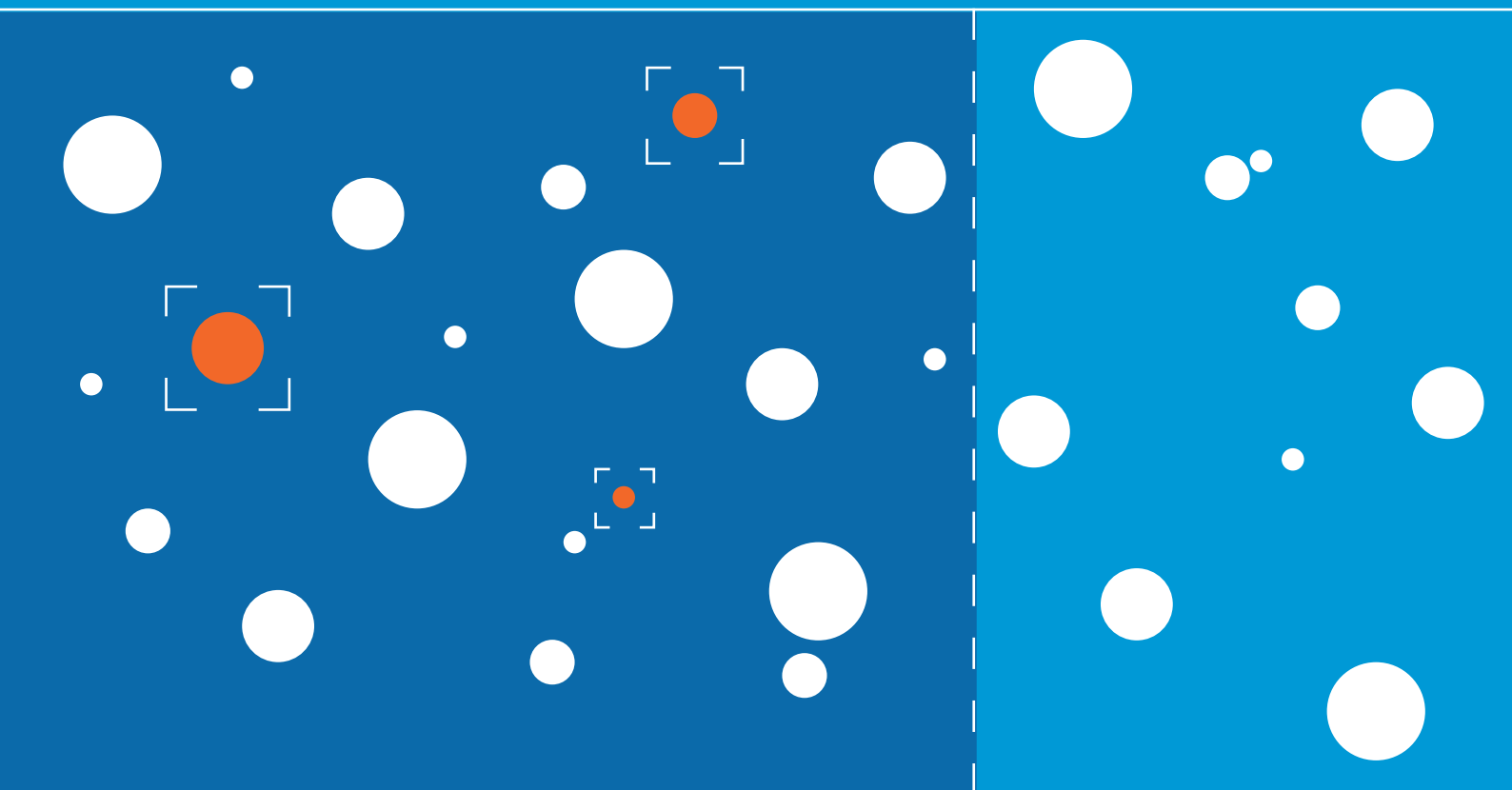


Guidelines for HIV post-exposure prophylaxis

Web Annex B. Systematic review findings and GRADE tables

Caitlin Kennedy, Rahel Dawit and Ping Teresa Yeh



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HIV post-exposure prophylaxis (PEP) in community settings and by lay health workers or through task sharing: a systematic review of effectiveness, case studies, values and preferences, and costs

Caitlin Kennedy, Rahel Dawit and Ping Teresa Yeh

2 November 2023

Background

Post-exposure prophylaxis (PEP) for HIV prevention is recommended by the World Health Organization (WHO) (1). In many contexts, however, PEP is not widely available outside of hospital settings for occupational or sexual assault exposure, and uptake remains low. PEP has the potential to be more strategically used for HIV prevention through expanded access in community settings or through provision by trained lay health workers as well as professional health workers. For example, in the United States of America (USA), laws have been recently changed to allow pharmacists to initiate or prescribe PEP in New York (as of 2017) (2) and California (as of 2019) (3), along with 10 other states (2). Pharmacies have been proposed as a promising location for PEP prescribing in South Africa (4). However, the scientific evidence to inform programmatic and policy decisions in these areas has not been synthesized.

We sought to conduct a systematic review of effectiveness, case studies, values and preferences, and cost data related to the provision of HIV PEP in community settings and by trained lay health workers or through task sharing.

Methods

Effectiveness review

The effectiveness review covered two complementary questions framed using the PICO approach (population, intervention, comparator, outcomes). The first PICO question asked, should PEP be offered in community settings? The second PICO question asked, should PEP be offered by lay health workers or through task sharing?

PICO 1: PEP in community settings

Population: Individuals eligible for PEP

Intervention: Availability of PEP in community settings (defined as non-health care settings)

Comparator: No availability of PEP, or availability only in health care settings

Outcomes:

1. Quality of PEP services offered (for example, adherence to country or international guidelines)
2. Uptake of PEP
3. Timeliness of PEP uptake (time since exposure)
4. Completion of PEP
5. HIV acquisition
6. Linkage to or uptake of appropriate additional services (for example, PrEP, ART)
7. Adverse events (that is, coercion, intimate partner violence, self-harm, psychosocial issues, stigma/discrimination).

PICO 2: PEP offered by lay health workers or through task sharing

Intervention: PEP provided by trained lay health workers, as defined by the WHO definition of a lay health worker as “any health worker who performs functions related to health care delivery; was trained in some way in the context of the intervention, but has received no formal professional or paraprofessional certificate or tertiary education degree” (5). We also included studies that focused on task sharing, where PEP was provided by a lower-level provider.

Comparator: PEP provided by trained health workers/higher level health workers, or no provision of PEP.

Outcomes:

1. Quality of PEP services offered (for example, adherence to country or international guidelines)
2. Uptake of PEP
3. Timeliness of PEP uptake (time since exposure)
4. Completion of PEP
5. HIV acquisition
6. Linkage to or uptake of appropriate additional services (for example, PrEP, ART)
7. Adverse events (for example, coercion, intimate partner violence, self-harm, psychosocial issues, stigma/discrimination).

Although these PICO questions focused on PEP in community settings and PEP offered by lay health workers, we also included any comparison of two different settings where PEP is available (for example, hospital versus primary health care setting, hospital versus pharmacy, etc.) as well as included studies that compared PEP offered by different cadres of health workers (for example, physicians versus pharmacists, pharmacists versus nurses, etc.).

Inclusion and exclusion criteria

Studies were included in the PICO reviews if they met the following criteria:

- 1) Study population included individuals eligible for PEP (according to country or international guidelines).
- 2) Study design was a randomized trial or comparative observational study (including non-randomized quasi-experimental studies) that compared people who received the intervention described in the PICO question with those who received an intervention described in the PICO comparison group.
- 3) Measured one or more of the outcomes of interest.
- 4) Published in a peer-reviewed journal or as a conference abstract between January 1, 2012 (the year when antiretroviral drugs for treatment were recommended in community settings by WHO) through the search date of October 16, 2023.

If studies combined both PEP and pre-exposure prophylaxis (PrEP), or they combined both PEP and treatment as prevention (TasP) for the purposes of assessing “biomedical HIV prevention” more generally, we reported only PEP-specific findings if findings were disaggregated; if findings were not disaggregated, we reported findings but made clear that these were for interventions beyond just PEP. Studies from any geographic region and any language were included. For studies published in languages other than English, we reviewed the English-language abstract, if one was available, or a Google Translate version of the abstract if it was in a language not spoken by the study team.

Search strategy

We used a single search strategy to identify articles for the PICO, values and preferences, and cost reviews that searches titles and abstracts that include terms for “PEP” and “HIV”. This broad search was intended to maximize sensitivity. We searched four online databases (CINAHL, PsycINFO, PubMed and EMBASE) for relevant peer-reviewed publications. The full search strategy for each database was as follows:

PubMed: (HIV[Title/Abstract] or HIV[MeSH]) AND (post-exposure prophylaxis [MeSH] OR “postexposure prophylaxis”[Title/Abstract] OR “post-exposure prophylaxis”[Title/Abstract] OR “post exposure prophylaxis”[Title/Abstract] OR PEP[Title/Abstract]))

CINAHL: (MH HIV OR AB HIV OR TI HIV) AND (MH “post-exposure prophylaxis” OR AB “postexposure prophylaxis” OR AB “post-exposure prophylaxis” OR AB “post exposure prophylaxis” OR AB PEP OR TI “postexposure prophylaxis” OR TI “post-exposure prophylaxis” OR TI “post exposure prophylaxis” OR TI PEP)

PsycINFO: (MH HIV OR AB HIV OR TI HIV) AND (MH “post-exposure prophylaxis” OR AB “postexposure prophylaxis” OR AB “post-exposure prophylaxis” OR AB “post exposure prophylaxis” OR AB PEP OR TI “postexposure prophylaxis” OR TI “post-exposure prophylaxis” OR TI “post exposure prophylaxis” OR TI PEP)

EMBASE: ('postexposure prophylaxis':ab,ti OR 'post-exposure prophylaxis':ab,ti OR 'post exposure prophylaxis':ab,ti OR pep:ab,ti) AND hiv:ab,ti)

In addition to the online database search, we also searched conference abstracts from the CROI, IAS and IAC conferences. We reviewed the reference lists of several previously conducted reviews (6-8) and of all included studies. Finally, we asked Guideline Development Group members to identify potentially relevant articles we may have missed.

Screening process

Titles, abstracts, citation information and descriptor terms of citations identified through the search strategy were screened by a member of the review team (PTY, RD, CK). Full-text articles were obtained of all selected abstracts, and two reviewers (CK, RD) independently assessed all full-text articles for eligibility to determine final study selection. Differences were resolved through consensus.

Data extraction, management and analysis

Data were extracted independently by two reviewers (CK, RD) using standardized data extraction forms in Excel. Differences in data extraction were resolved through consensus. From each study we gathered information on citation information (author, year, title, journal, language of article), location, study population description, sample size, study design, intervention summary, comparator description (when applicable) and study outcomes. Risk of bias at the study level was assessed using the Cochrane Collaboration risk of bias tool for randomized trials (9) and the Evidence Project risk of bias tool (10) for comparative observational studies. We also assessed risk of bias at the level of individual outcomes using GRADE evidence profiles.

Data were analysed according to coding categories and outcomes. Where there were multiple studies reporting the same outcome for the same intervention-comparator comparison for the same population, we planned to conduct meta-analysis using random-effects models to combine risk ratios (for dichotomous outcomes) or either the mean difference (MD) or standardized mean difference (SMD) (for continuous outcomes) separately for RCTs versus comparative observational studies. We planned to explore heterogeneity in meta-analyses using the I-squared statistic and interpreted either through qualitative examination of forest plots or considering the Cochrane Handbook's rule of thumb (0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity). We planned to explore explanations for heterogeneity and create funnel plots to examine the potential for publication bias when there were a sufficient number of studies.

We planned to present all outcomes stratified by intervention type and population. In particular, we planned to stratify data by the following considerations:

- PEP following sexual assault versus PEP following consensual sex versus PEP following other situations (for example, occupational exposure, needle/syringe use);
- PEP for adolescents and young people ages <18 years versus PEP for adults;
- PEP for key populations (including criminalized populations) versus PEP for other populations.

Findings for the PICO reviews were summarized in GRADE Evidence Profile tables (11) (see below).

Case studies review

The same search terms were used to search and screen for studies to be included in the case studies review. Studies were included in this review if they presented primary data examining the implementation of a programme for community-based PEP or provision of PEP by trained lay health workers/task-sharing and provided information on implementation characteristics or outcomes from this programme, but the data did not meet the pre/post or multi-arm criteria for the PICO reviews. These studies could be qualitative or quantitative in nature but had to present primary data collection; think pieces and review articles were not included. These could include studies that examined barriers and facilitators to PEP in community settings or by lay health workers/task sharing or studies that provided non-comparative outcome data on the PICO outcomes listed above. Case study articles were summarized qualitatively and were organized by study design and methodology, location, intervention and population. We looked for data on the same stratifications as noted for the PICO questions above.

Values and preferences review

The same search terms were used to search and screen for studies to be included in the values and preferences review. Studies were included in this review if they presented primary data examining the values and preferences of potential beneficiaries, communities, health workers and other stakeholders for community-based PEP or provision of PEP by trained lay health workers/task sharing. These studies could be qualitative or quantitative in nature but had to present primary data collection; think pieces and review articles were not included. This literature could include studies examining the acceptability of various intervention options covered in the PICO questions above and service delivery preferences, among others. Studies that reported only uptake of PEP or awareness of PEP as a proxy for preferences were not included. Values and preferences literature was summarized qualitatively and was organized by study design and methodology, location, intervention and population. We looked for data on the same stratifications as noted for the PICO questions above.

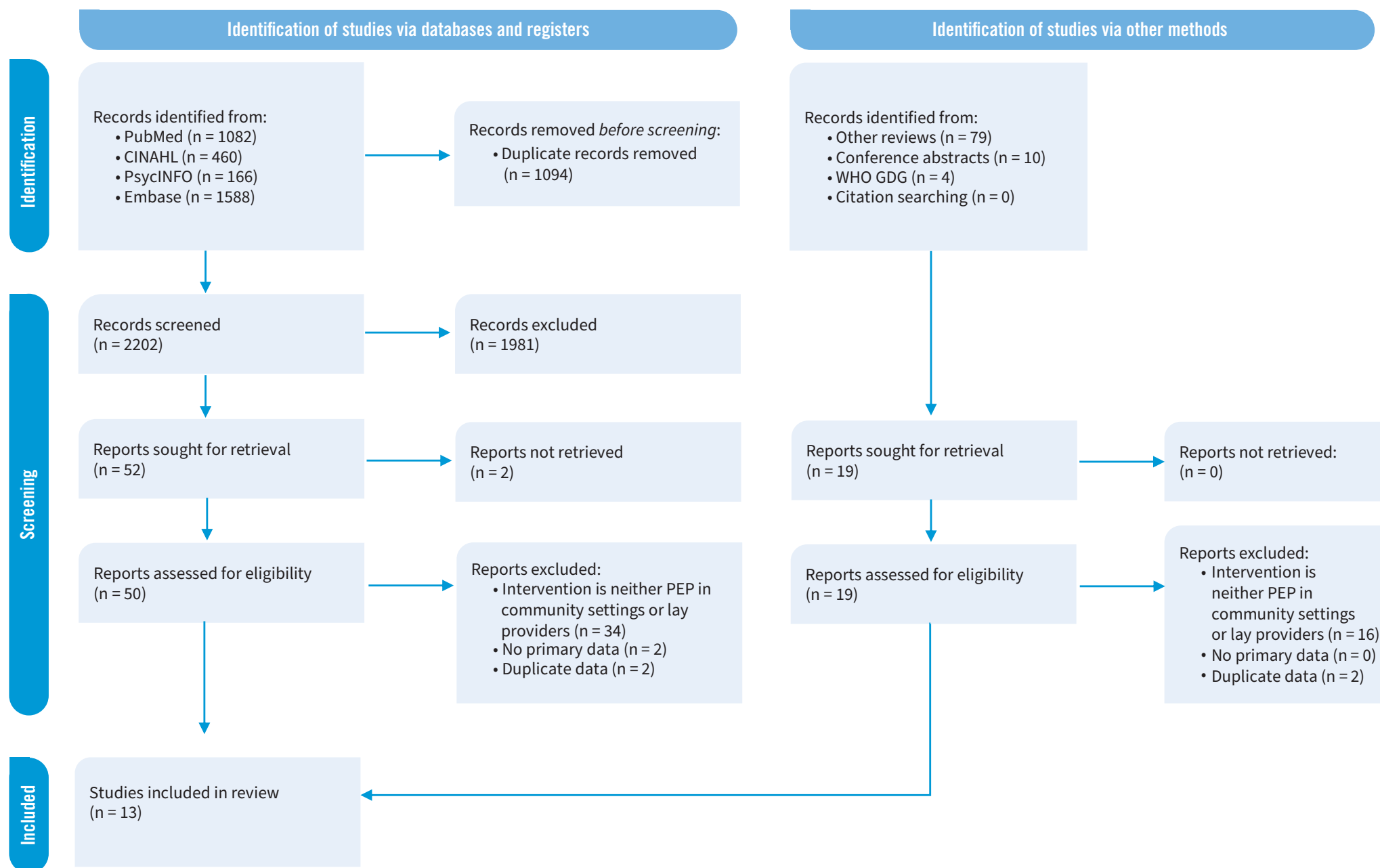
Cost and resource needs

The same search terms were used to search and screen for studies to be included in the cost review. Studies were included in this review if they presented primary data comparing costing, cost-effectiveness, cost-utility or cost-benefit of community-based PEP or provision of PEP by trained lay health workers/task sharing. Cost studies were summarized qualitatively. We planned to classify cost literature into four categories (health sector costs, other sector costs, patient/family costs and productivity impacts) and within each category organize the data by study design/methodology, location, intervention and population. We looked for data on the same stratifications as noted for the PICO questions above.

Results

Fig. 1 presents a PRISMA diagram showing the disposition of citations through the search and screening process (12). The database search identified 2202 unique citations, of which 50 were assessed for full-text review. Additional search methods identified 93 additional citations. Overall, 13 studies met our inclusion criteria: three effectiveness studies, five case studies, five values and preferences studies (one of which was reported in the same abstract as a case study), and two cost studies (one of which was reported in the same abstract as an effectiveness study). Given the small number of studies and diversity of outcomes, we did not conduct meta-analysis or stratify data. Below, we present results summarized by the effectiveness, case study, values and preferences and cost reviews and separated by the two interventions of interest (PEP offered in community settings and PEP offered by lay health workers/task-sharing).

Fig. 1. PRISMA flowchart showing selection of articles into the systematic review



Effectiveness review

One study met the inclusion criteria for the effectiveness review of PEP offered in community settings (Table 1) (13). The SEARCH SAPPHIRE study, conducted in Kenya and Uganda, offered PEP as part of a “dynamic choice” model of HIV prevention options, which also included PrEP and condoms. The dynamic choice model was delivered in three different settings: in antenatal care (ANC) settings and outpatient departments (OPD), where PEP was offered by clinical officers and nurses, and in community settings, where PEP was offered by community health workers who facilitated the intervention by clinical officers from the local health centre. At intervention visits during weeks 4, 12 and 24, participants were asked to select a choice of HIV prevention option (PrEP, PEP, condoms only and no selection), HIV testing modality (oral self-test or clinician-administered rapid antibody) and preferred location for next visit (clinic versus out-of-facility). At week 24 PEP use and HIV risk (report of sexual partners with HIV or unknown status and/or self-identification as being at risk) for each of the prior six calendar months were assessed via a structured survey. This observational study started as low certainty of evidence in GRADE and was downgraded to very low certainty evidence due to risk of bias, because populations accessing the intervention through community, ANC and OPD settings are substantially different on demographic factors (for example, gender, age, pregnancy status) and due to inconsistency, because it was a single study (Table 2). The study measured the outcome of uptake of PEP over 24 weeks of follow-up and found that the initial choice of PEP for HIV prevention was highest in the community setting (46%) compared with the OPD and ANC settings (9% and 1%, respectively). Selection of PEP over the follow-up study visits remained highest in the community setting over time (23% at week 24); in the ANC and OPD settings, only 3% and 11%, respectively, ever selected PEP.

Table 1. Description of studies included in the effectiveness review

Study	Country	PEP intervention	Setting	Study design	Sample size
PEP offered in community settings					
Kabami et al., 2022 (13)	Kenya and Uganda	PEP offered by CHWs in community settings as part of a dynamic HIV prevention choice model settings	Community	Non-randomized trial	612 community participants
PEP offered by lay health workers/task-sharing					
Kabami et al., 2022 (13)	Described above				
Grossman et al., 2020 (14)	USA	Pharmacist prescribing PEP	Infectious disease clinic after referral from emergency department	Retrospective chart review before/after intervention	24 PEP clients
Lowrey et al., 2020 (15)	USA	Pharmacist prescribing free PEP, providing education, and conducting a follow-up call	Emergency department	Retrospective chart review before/after intervention	369 PEP clients

CHWs: community health workers; PEP: post-exposure prophylaxis

Table 2. GRADE evidence profile for PICO 1: PEP offered in community settings

Certainty assessment							Nº of patients		Effect		Certainty	Importance
Nº of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Availability of PEP in community settings	No availability	Relative (95% CI)	Absolute (95% CI)		
Uptake of PEP (follow-up: mean 24 weeks; assessed with: Selection of PEP or other biomedical prevention options [PrEP, condoms, none])												
1 ^{1,a}	observational studies	serious ^b	serious ^c	not serious	not serious	none	The initial choice of PEP for HIV prevention was highest in the community setting (46%) compared with the OPD and ANC settings (9% and 1%, respectively). Selection of PEP remained highest in the community setting over time (23% at week 24); in the ANC and OPD settings, only 3% and 11%, respectively, ever selected PEP.			⊕○○○ Very low	CRITICAL	

CI: confidence interval

Explanations

- ^a Intervention: Dynamic choice of biomedical HIV prevention comparing community delivery with delivery through OPD and ANC settings.
- ^b Risk of bias: Downgraded because populations accessing the intervention through community, OPD and ANC settings are substantially different on demographic factors (for example, gender, age, pregnancy status).
- ^c Inconsistency: Downgraded for being a single study.

References

- Kabami J, Kakande E, Chamie G, Balzer LB, Petersen ML, Camlin CS et al. Uptake of a patient-centred dynamic choice model for HIV prevention in rural Kenya and Uganda: SEARCH SAPHIRE study. *J Int AIDS Soc.* 2023;26 Suppl 1:e26121. doi:10.1002/jia2.26121

The SEARCH SAPHIRE study was also considered to meet the inclusion criteria for the effectiveness review of PEP offered by lay health workers. In addition, two additional conference abstracts reported on studies conducted in the United States of America (Table 1).

The first study conducted a retrospective chart review of PEP cases before and after a programme to allow a pharmacist in an infectious disease clinic to prescribe PEP following referral from an emergency department; previously, PEP cases were seen in the infectious disease clinic without pharmacist involvement (14). This observational study started as low certainty evidence and was downgraded to very low certainty for having a small sample size (n=24 PEP cases across both arms) (Table 3). This study measured two primary outcomes: PEP uptake and PEP completion. After the intervention, 16/16 (100.0%) of eligible clients left the clinic with PEP, compared with 5/8 (62.5%) prior to the intervention. After the intervention, 42% of those who left the clinic with PEP completed the entire PEP course and came to a follow-up appointment, compared with 32% before the intervention (n's not reported).

The second study also conducted a retrospective chart review comparing before and after a programme that involved pharmacists in dispensing free PEP, providing patient education prior to discharge and conducting a follow-up phone call after three months for sexual assault victims in an emergency department (15). This observational study started as low certainty evidence and was downgraded to very low certainty evidence due to a small number of events (Table 3). The study measured the outcomes of PEP completion and HIV acquisition. PEP completion was 19.8% (n=55) in the pharmacist-delivered interventions. 4.3% (n=4) before (the total sample size was 369, but the sample size in each arm is not reported). There were two documented cases of HIV seroconversion before the intervention and none afterwards.

No studies included in the effectiveness review measured the outcomes of quality of PEP services offered, timeliness of PEP uptake, linkage to or uptake of appropriate additional services, or adverse events.

Table 3. GRADE evidence profile for PICO 2: PEP offered by lay health workers/task-sharing

Certainty assessment							Nº of patients		Effect		Certainty	Importance
Nº of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	PEP offered by lay health workers/ task-sharing	No availability	Relative (95% CI)	Absolute (95% CI)		
Uptake of PEP (follow-up: mean 24 weeks; assessed with: Selection of PEP compared with other biomedical prevention options (PrEP, condoms, none))												
1 ^{1,a}	observational studies	serious ^b	serious ^c	not serious	not serious	none	The initial choice of PEP for HIV prevention was highest in the community setting (46%) compared with the OPD and ANC settings (9% and 1%, respectively). Selection of PEP remained highest in the community setting over time (23% at week 24); in the ANC and OPD settings, only 3% and 11%, respectively, ever selected PEP.				⊕○○○ Very low	CRITICAL
Uptake of PEP (assessed with: Receiving PEP prior to leaving the medical centre)												
1 ^{2,d}	observational studies	not serious	serious ^c	not serious	serious ^e	none	16/16 (100.0%)	5/8 (62.5%)	not estimable		⊕○○○ Very low	CRITICAL
Completion of PEP (assessed with: Completed entire PEP course and came to follow-up appointment)												
1 ^{2,d}	observational studies	not serious	serious ^c	not serious	serious ^e	none	42% in intervention versus 32% in comparison arm; n's not reported.				⊕○○○ Very low	CRITICAL
Completion of PEP (assessed with: Presenting for follow-up within 6 months)												
1 ^{3,f}	observational studies	not serious	serious ^c	not serious	not serious	none	19.8% (n=55) in intervention versus 4.3% (n=4) in comparison arm; total sample size was 369, but sample size in each arm not reported.				⊕○○○ Very low	CRITICAL
HIV acquisition (assessed with: Documented seroconversion within 6 months)												
1 ^{3,f}	observational studies	not serious	serious ^c	not serious	serious ^g	none	0 seroconversions in the intervention group versus 2 in the comparison group; sample size in each arm not reported.				⊕○○○ Very low	CRITICAL

CI: confidence interval

Explanations^a Intervention: Dynamic choice of biomedical HIV prevention comparing community delivery versus delivery through OPD and ANC settings.^b Risk of bias: Downgraded because populations accessing the intervention through community, outpatient department (OPD) and antenatal care (ANC) settings are substantially different on demographic factors (for example, gender, age, pregnancy status).^c Inconsistency: Downgraded for being a single study.

- ^d Intervention: Adding a pharmacist to deliver PEP in an infectious disease clinic for cases referred from emergency department compared with prior to pharmacist involvement.
- ^e Imprecision: Downgraded for small sample size (n=24 across both arms).
- ^f Intervention: Pharmacist involvement in dispensing free PEP and patient education plus follow-up phone call after three months for sexual assault victims in an emergency department compared with prior to these interventions.
- ^g Imprecision: Downgraded for small number of events.

References

1. Kabami J, Kakande E, Chamie G, Balzer LB, Petersen ML, Camlin CS et al. Uptake of a patient-centred dynamic choice model for HIV prevention in rural Kenya and Uganda: SEARCH SAPPHIRE study. *J Int AIDS Soc.* 2023;26 Suppl 1:e26121. [doi:10.1002/jia2.26121](https://doi.org/10.1002/jia2.26121)
2. Grossman, K.H. Improving patient access to HIV post-exposure prophylaxis with pharmacist involvement. *Open Forum Infect Dis.* 2020;7:S522.
3. Lowrey, K., Kaucher, K., Dawson, J., Gilliam, E.. Improvement in HIV screening follow-up with emergency department pharmacist dispensing of post-exposure prophylaxis. *J Am Coll Clin Pharm.* 2020;3:1573-4. [doi:10.1002/jac5.1351](https://doi.org/10.1002/jac5.1351)

Case studies review

Three studies were included in the case studies review of PEP offered in community settings, two from Africa and one from China (Table 4). The first study trained health workers from government clinics in Kenya and Uganda on PEP with an option for out-of-facility, community-based medication delivery (16). Among 124 persons who sought PEP, 85% met the definition of PEP completion, and there were no serious adverse events or HIV seroconversions reported. Overall, 12% of visits were conducted at out-of-facility community-based sites; 35% of participants had at least one out-of-facility visit. The second study examined police initiation of PEP at police stations for sexual assault survivors in Zambia (17). Of 207 cases of sexual assault, about half were eligible for PEP (n=104), but only 25% of these (n = 26) were initiated on PEP by the police. The authors noted that less than half of eligible cases (n=49) presented during official police working hours, and 33% of eligible survivors who reported during official working hours received PEP, compared with 18% of those who reported on nights or weekends. No adverse events were reported. The third study examined a web-based platform for delivering PEP in China (18). Of 539 PEP users who responded to the survey, nearly all (99%) started PEP within 72 hours of exposure, and there were no HIV seroconversions reported.

Two studies were included in the case studies review of PEP offered by lay health workers/task-sharing (Table 4). The first study evaluated PEP delivery in 12 private pharmacies in Kisumu and Kiambu counties in Kenya (19). Of 989 clients screened over a six-month period for PEP, PrEP or STI testing, 173 clients were initiated on PEP, and 18% (32/173) of these transitioned to PrEP upon PEP completion. The second study trained 14 nurse prescribers to offer PEP in nurse-delivered clinics in the UK, compared with the usual offer through a central sexual health hub clinic and emergency department, and compared their outcomes with a sample of PEP prescribed by doctors over the same period (20). In the six months after training, 27 PEP assessments were completed by nine nurses across six satellite clinics. Nurses became the second highest prescriber group of PEP. Of 27 patient assessments, 19 received PEP, with 18/19 of those prescribing decisions meeting local prescribing criteria. The authors reported that the nurses' prescribing "was comparable, if not better, than the doctors' prescribing", although no data were shown.

Table 4. Description of studies included in the case studies review

Study	Country	PEP setting/ intervention	Population	Sample size	Results
PEP offered in community settings					
Ayieko et al., 2021 (16)	Kenya and Uganda	Community-based delivery	General population	124 PEP initiations	124 persons sought PEP; 85% completed PEP, and there were no HIV seroconversions. 12% of all visits were conducted at out-of-facility community-based sites; 35% of participants had ≥1 out-of-facility visit. - No serious adverse events were reported.
Zama et al., 2015 (17)	Zambia	Police stations	Sexual assault survivors	207 cases of sexual assault	- About half of cases were PEP-eligible, and 25% of these were initiated on PEP by the police. - Less than half of eligible cases presented during working hours, but eligible cases were more likely to receive PEP if during working hours (33%) than if during non-working hours (18%).
Shan et al., 2023 (18)	China	Internet-based service	Mostly men who have sex with men	539 PEP users	Of 539 PEP users who responded to the survey, nearly all (99%) started PEP within 72 hours of exposure, and there were no HIV seroconversions reported.
PEP offered by lay health workers/task-sharing					
Roche et al., 2023 (19)	Kenya	Private pharmacies	General population	989 clients	173 clients were initiated on PEP, and 18% (32/173) of these transitioned to PrEP upon PEP completion.
Mensforth et al., 2018 (20)	UK	Nurse-delivered clinics	Not reported	27 PEP assessments	- After training, nurses became the highest provider of PEP. - Of 19 PEP prescribing decisions, 18 met local prescribing criteria. - Nurses' prescribing was reported as "comparable" to doctors' prescribing.

Values and preferences review

One study was identified for the values and preferences review of PEP offered in community settings (Table 5). This study was a cross-sectional survey of 342 sexual and gender minorities visiting collective sex venues in New York City, USA (21). In open-text survey responses, participants expressed interest in such venues providing a range of free HIV and STI prevention services, including PEP. Although results were not separated for PEP services, participants felt services could be delivered in an acceptable way, although potential barriers included privacy concerns, preexisting access to health services, an emphasis on personal responsibility and negative reactions to the presence of service providers.

Four studies were identified for the values and preferences review of PEP offered by lay health workers/task-sharing (Table 5). Two of these were online, cross-sectional surveys of non-randomly selected PEP providers. The first study used an online survey of 214 nurses in Ontario, Canada to assess perspectives on allowing nurses to dispense PEP. Overall, 76.9% of participants indicated they would be supportive of nurse-led PEP under medical directives (22). The second study was a multi-country, mixed-methods study to examine values and preferences around PEP to inform prior WHO guidelines (23). The online survey component was completed by 306 participants from five countries: Armenia (n=16), Kenya (n=15), Lesotho (n=16), South Africa (n=90) and the USA (n=51). Of these providers, 65.5% (n=110) disagreed that 28-day prescribing should be done only by HIV specialists, and 74.1% (n=126) agreed that they could allow non-HIV specialists to start PEP safely. The third study conducted semi-structured qualitative interviews with staff at PEP-prescribing pharmacies in the San Francisco Bay area, USA (24). Of seven interview participants, all felt the California state bill that allowed pharmacists to dispense PEP was a valuable expansion of pharmacist functions, and two “indicated that furnishing PEP was a public health duty.” Finally, the fourth study, also included in the case study review, evaluated a model of PEP delivery (along with PrEP and HIV testing) in private pharmacies in Kenya (19). Although PEP was not separated from PrEP in the analysis, acceptability was generally high. The majority (70–100%) of clients and providers reported that they liked getting/delivering PrEP/PEP at the pharmacy and that getting/delivering PrEP/PEP at the pharmacy was not hard.

Table 5. Description of studies included in the values and preferences review

Study	Country	Participants	Study design	Sample size	Results
PEP offered in community settings					
Cai et al., 2023 (21)	USA	Sexual and gender minorities	Cross-sectional survey	342	Participants expressed interest in receiving PEP, among other HIV and STI prevention services, at collective sex venues.
PEP offered by lay health workers/task-sharing					
Clifford-Rashotte et al., 2018 (22)	Canada	Nurses	Cross-sectional survey	214	76.9% support nurse-led PEP under medical directives.
Beanland et al., 2015 (23)	Multi-country (Armenia, Kenya, Lesotho, South Africa, USA)	Health workers	Cross-sectional survey	306	65.5% disagree that PEP should be prescribed only by HIV specialists. 74.1% said non-HIV specialists could start PEP safely.
Bellman et al., 2022 (24)	USA	Pharmacists	Semi-structured qualitative interviews	7	All participants felt the state bill allowing pharmacists to provide PEP was a valuable expansion of pharmacist functions, and two “indicated that furnishing PEP was a public health duty”.
Roche et al., 2023 (19)	Kenya	PEP clients and pharmacy providers	Cross-sectional survey	Not reported	Acceptability of pharmacy offer of PEP was high, with the majority (70–100%) of clients and providers reporting that they liked getting/delivering PrEP/PEP at the pharmacy and that getting/delivering PrEP/PEP at the pharmacy was not hard.

Cost and resource needs

Two studies were included in the cost and resources review; one for PEP offered in community settings, and one for PEP offered by lay health workers/task-sharing (Table 6). The first study used mathematical modelling to examine a range of scenarios around wider PEP availability in communities in west, east, central and southern Africa (25). This study estimated that three months of PEP use in Africa, including a 20% additional supply chain cost to cover distribution, would cost US\$ 16.20. Overall costs were lower with community PEP than with no community PEP in 92% of setting scenarios, with US\$ 18.0 million (14% of the overall HIV budget of US\$ 127.8 million per year) savings per year over 50 years as a result of fewer people requiring ART and lower ART-related clinic visits over the long-term. Models suggested that community PEP was cost-effective in 90% of setting scenarios and cost-saving (with disability-adjusted life-years averted) in 58% of scenarios. When examining only setting scenarios in which there was lower uptake of community PEP, it was found to be cost-effective in 92% of setting scenarios. The second study, also included in the effectiveness review, assessed cost savings for PEP offered by pharmacists in the USA (14). This study found clients' average out-of-pocket costs for PEP were US\$ 2.25–7.30 after the pharmacist intervention, compared with US\$ 475.00–3733.40 before the pharmacist intervention.

Table 6. Description of studies included in the costs review.

Study	Country	Setting	Population	Sample size	Outcomes
PEP offered in community settings					
Phillips et al., 2023 (25)	Multiple countries in Africa	Community-based availability of PEP	General population	Modelling study	- Community PEP was cost-effective in 90% of setting scenarios and cost-saving (with disability-adjusted life-years averted) in 58% of scenarios. - When only examining setting scenarios in which there was lower uptake of community PEP, it was cost-effective in 92% of setting scenarios.
PEP offered by lay health workers/task-sharing					
Grossman et al., 2020 (14)	USA	Pharmacists prescribing PEP	Patients referred from an emergency department to an infectious disease clinic	24	Average out-of-pocket costs for patients were US\$ 2.25, with a maximum total cost of US\$ 7.30, as opposed to US\$ 475 and US\$ 3733.40 prior to pharmacist engagement. The involvement of pharmacists resulted in significant cost savings for patients receiving PEP.

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